

Parameter Studies

Parameter Study – Constant A:

Aim:

To determine the relationship between the constant A and the final y displacement of the curling stone.

To compare the final y displacement of clockwise and anti-clockwise rotations of the curling stone.

Prediction:

The value of the final y displacement will be directly proportional to the constant, A, and the line of best fit will have a y intercept of 0.

A clockwise rotation will produce a negative y displacement while an anti-clockwise rotation will produce a positive y displacement.

The clockwise and anti-clockwise rotation should produce symmetrical results.

Setup:

Parameter	Constant, A
Minimum	0
Maximum	0.25
Interval	0.01
Number Of Values	26
Runs Per Value	50
Total Number Of Runs	2,600
Other Parameters	Rotation direction changed, everything else is constant.

Results:

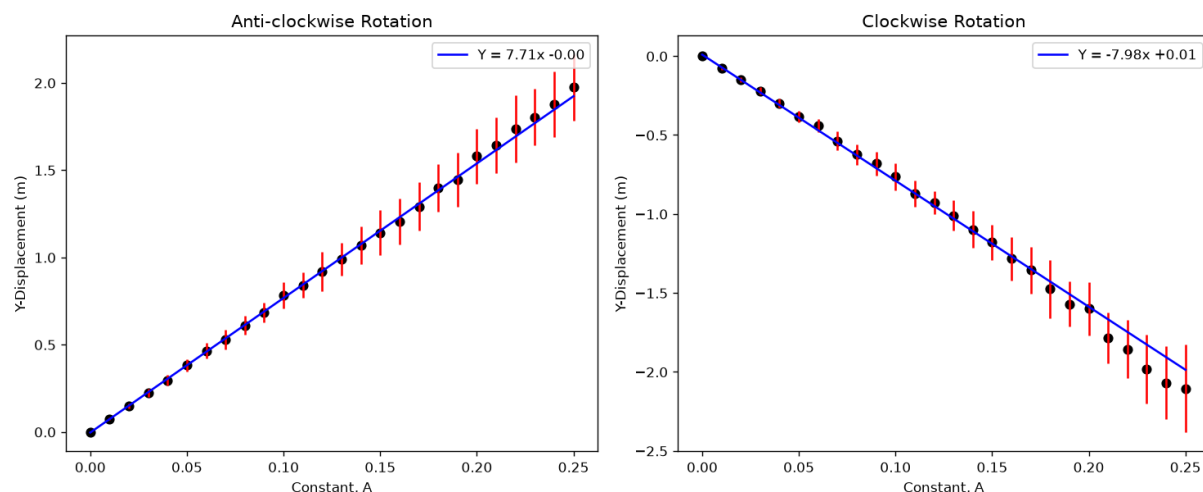


Figure 1: Two graphs showing the linear relationship between the constant A and the final y displacement of the curling stone, with standard deviation as error bars and a weighted line of best fit on each graph.

The clockwise graph has a gradient of 7.71m and a y-intercept of 0m while the anti-clockwise graph has a gradient of -7.98m and a y-intercept of 0.01m.

Total Runtime – 2347s

Conclusion:

The final y displacement of the curling stone is directly proportional to the constant A as shown in Figure 1, as predicted. It is also shown that the clockwise and anti-clockwise rotations give symmetrical results to each other, shown by closeness of the magnitude of the gradients of the line of best fit on each graph.

Parameter Study – Changing Mass:

Aim:

To investigate how the mass of the curling stone affects the final y displacement of the curling stone.

Prediction:

Increasing the mass of the curling stone will increase the final y displacement. The relationship between mass and final y displacement should be linear.

Setup:

Parameter	Mass (kg)
Minimum	17.24
Maximum	19.96
Interval	0.247
Number Of Values	11
Runs Per Value	50
Total Number Of Runs	1,650
Other Parameters	3 different values of A are used, everything else is constant.

Results:

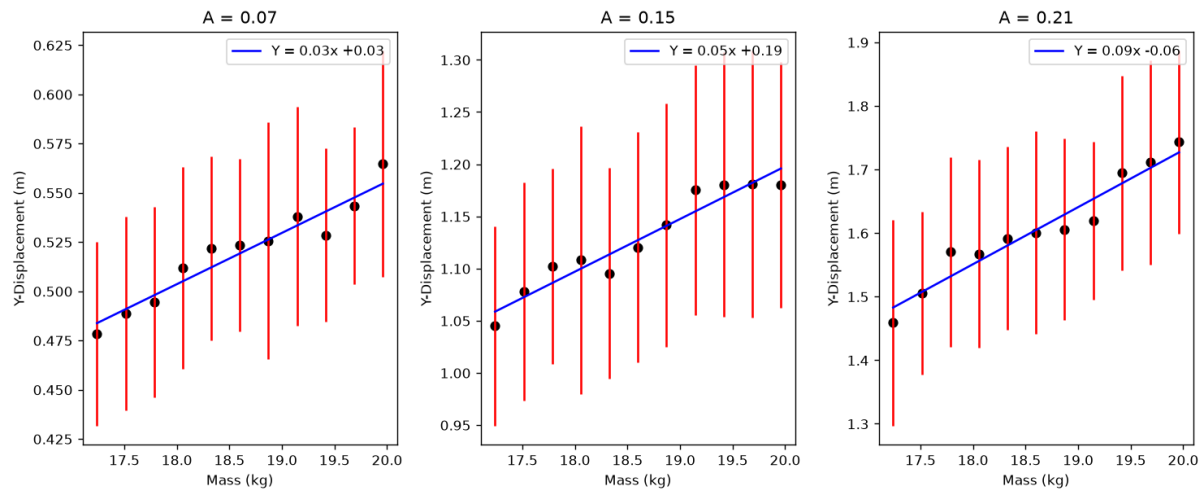


Figure 2: Graphs showing final y displacement of the curling stone against mass, for 3 different values of A. Each point is the average of 50 values, with a standard deviation and weighted lines of best fit.

Total Runtime – 1485s

Conclusion:

The final y displacement of the curling stone is directly proportional to the mass of the curling stone as shown in Figure 2, as predicted. Also, as the constant A is increase, the final y displacement increases as previously described.

Parameter Study – Initial Velocity Magnitude:

Aim:

Investigate the effect of the initial velocity on the final y displacement of the curling stone.

Prediction:

The graph of final y displacement against initial velocity should not be linear and have a one over initial velocity relationship.

Setup:

Parameter	Initial Velocity (m/s)
Minimum	2
Maximum	6
Interval	0.25
Number Of Values	17
Runs Per Value	50
Total Number Of Runs	850
Other Parameters	Kept constant

Results:

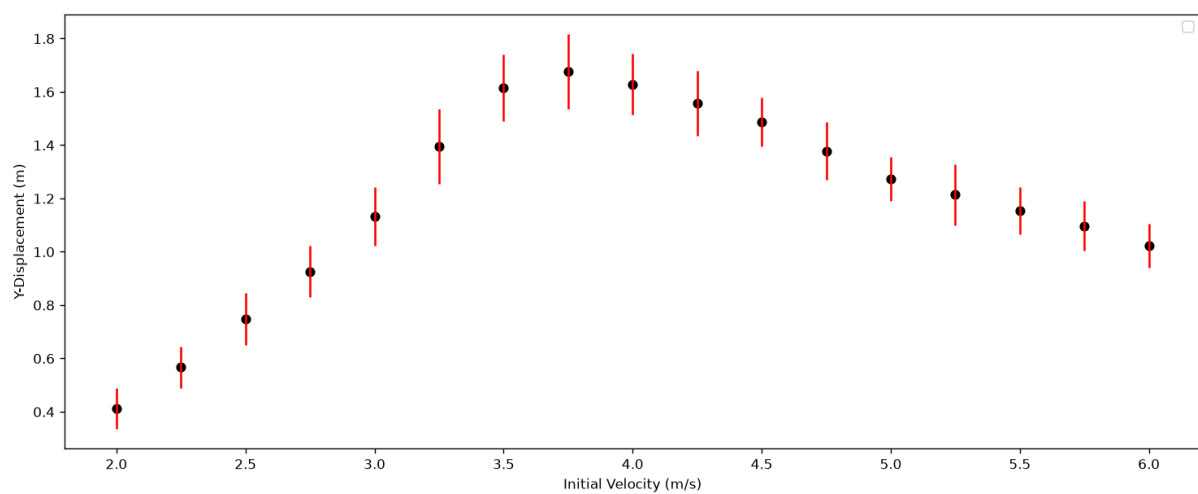


Figure 3: A graph of y displacement against initial velocity, each point is averaged from 50 results, with the standard deviation plotted as error bars.

Total Runtime – 783s

Conclusion:

The y displacement increases linearly between 2m/s and 3.7m/s before linearly decreasing at a slower rate. This is not as expected. This may be because the simulation was stopped before the curling stone reached the end of the run when the initial velocity was greater. Therefore, this should be repeated later ensuring that the curling stone reaches the end of its path before the y displacement is recorded.

Parameter Study – Running Band Width:

Aim:

To investigate the effects of the running band width on the final y displacement of the curling stone.

Prediction:

As the running band width increases, the final y displacement also increases.

Setup:

Parameter	Running Band Width (m)
Minimum	0.001
Maximum	0.011
Interval	0.001
Number Of Values	11
Runs Per Value	50
Total Number Of Runs	550
Other Parameters	Kept Constant

Results:

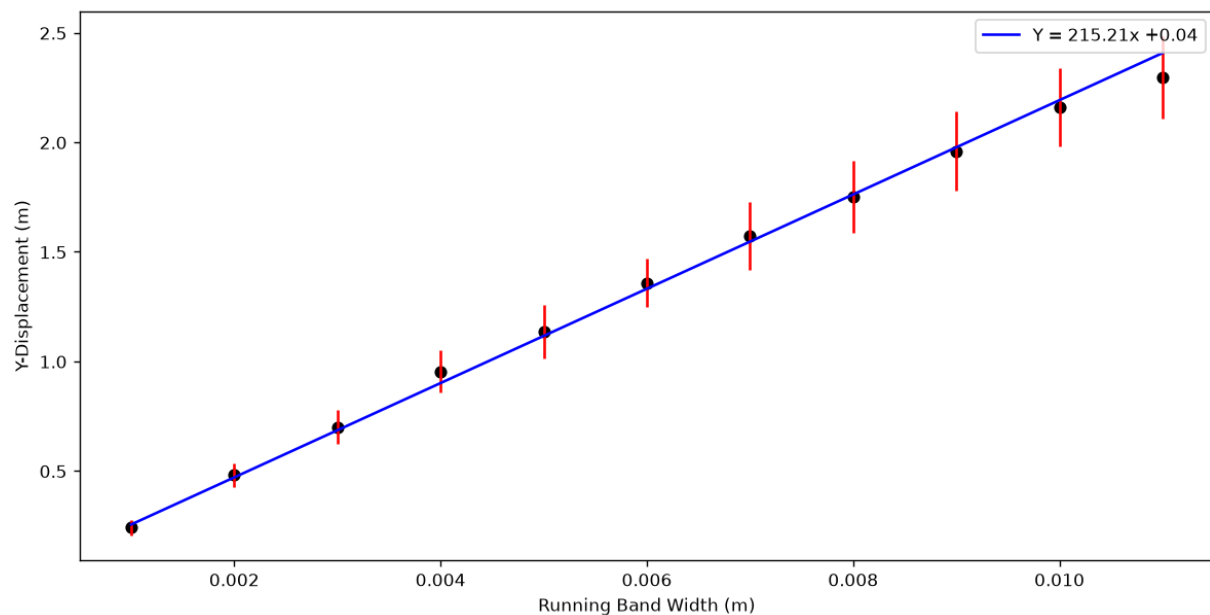


Figure 4: A graph of final y displacement of the curling stone against running band width, where each point is averaged from 50 results with the standard deviation plotted as error bars. A weighted line of best fit is plotted in blue.

The line of best fit has a gradient of 215.21 and a y-intercept of 0.04m.

Total Runtime – 702s

Conclusion:

The final y displacement of the curling stone is directly proportional to the width of the running band as shown in Figure 4 and as predicted. The line of best fit fits the data points very well and has a y-intercept close to 0 which is as expected.

Parameter Study – Running Band Radius:

Aim:

To investigate the effects of the running band radius on the final y displacement of the curling stone.

Prediction:

As the running band radius increases, the final y displacement also increases. This should be a linear relationship.

Setup:

Parameter	Running Band Radius (m)
Minimum	0.050
Maximum	0.080
Interval	0.025
Number Of Values	13
Runs Per Value	50
Total Number Of Runs	650
Other Parameters	Kept Constant

Results:

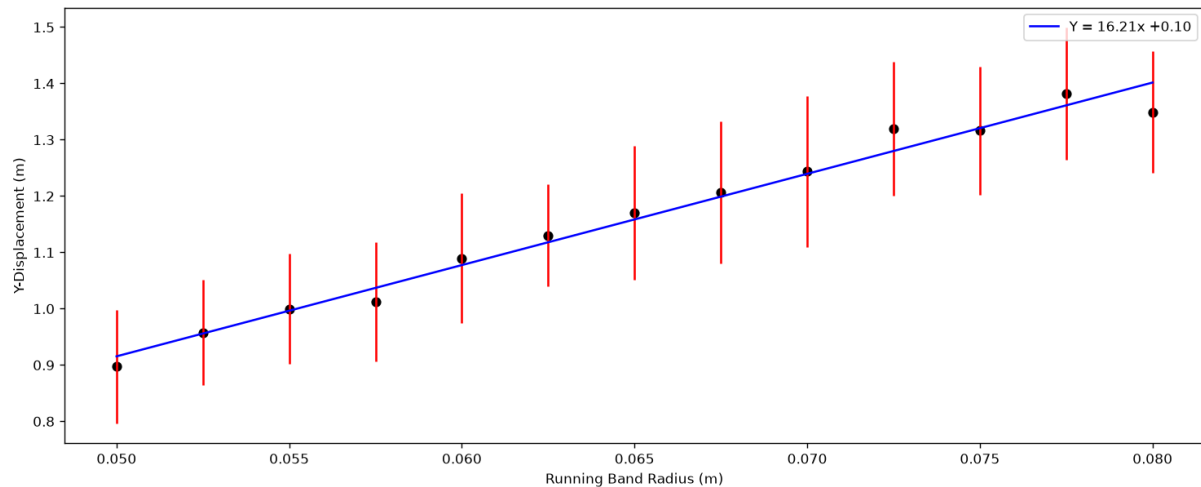


Figure 5: A graph of final y displacement of the curling stone against running band radius, each point is averaged from 50 results with the standard deviation plotted as error bars. A weighted line of best fit is plotted.

The line of best fit has a gradient of 16.21 and a y-intercept of 0.1m.

Total Runtime – 597s

Conclusion:

The final y displacement is directly proportional to the radius of the running band as predicted and also shown in Figure 5. The line of best fit fits the data points well and has a y-intercept of 0.1m. The expected y-intercept is 0 as a running band radius of 0 has 0 curl, however the y-intercept value on the graph is close.

Parameter Study – Pebble Density Factor:

Aim:

To investigate the effects of the pebble density factor on the final y displacement of the curling stone.

Prediction:

As the pebble density factor increases, the final y displacement should stay constant but the standard deviation should decrease as the pebble density factor increases.

Setup:

Parameter	Pebble Density Factor
Minimum	1
Maximum	8
Interval	0.5
Number Of Values	15
Runs Per Value	50
Total Number Of Runs	750
Other Parameters	Kept Constant

The pebble density factor is a multiplying factor, it multiplies the initial pebble density that is used. Therefore, a pebble density factor of 3 means there are 3 times as many pebbles in the same space.

Results:

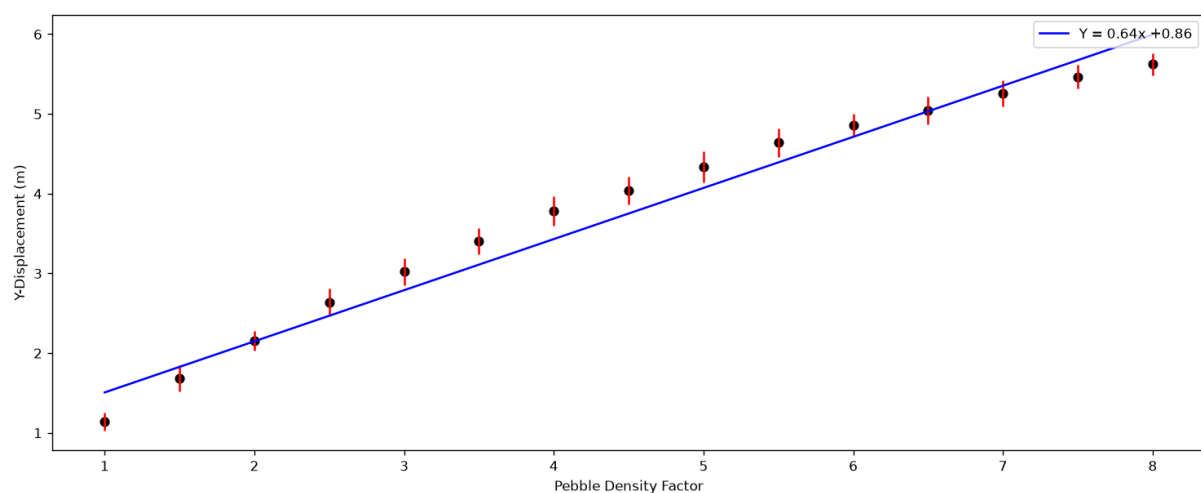


Figure 6: A graph of final y displacement of the curling stone against pebble density factor, where each point is averaged from 50 results, with the standard deviation plotted as error bars and a weighted line of best fit.

The line of best fit has a gradient of 0.64m and y-intercept of 0.86m

Total Runtime – 1717s

Conclusion:

For the current model, the final y displacement is directly proportional to the pebble density. This opposes the prediction, the final y displacement should be unaffected by the pebble density. Therefore, further work needs to be done to fix the model so that the final y displacement is not dependant on the pebble density.

Parameter Study – Average Pebble Radius:

Aim:

To investigate the effects of the average pebble radius on the final y displacement of the curling stone.

Prediction:

As the average pebble radius is increased, the final y displacement should also increase.

Setup:

Parameter	Average Pebble Radius (m)
Minimum	0.000325
Maximum	0.003900
Interval	0.000325
Number Of Values	12
Runs Per Value	50
Total Number Of Runs	600
Other Parameters	Kept Constant

Results:

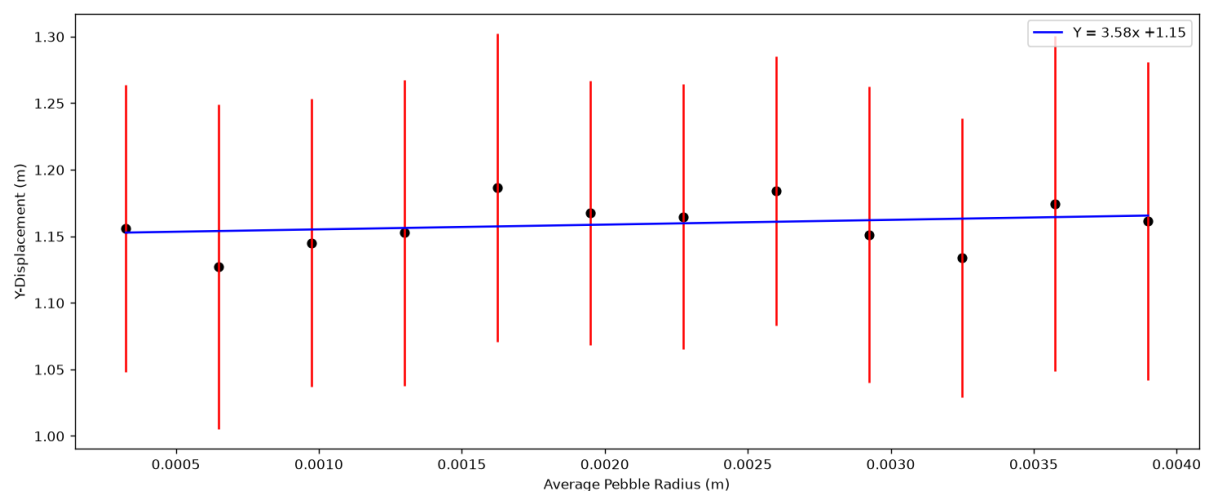


Figure 7: A graph of final y displacement of the curling stone against average pebble radius, each point is averaged from 50 results, with the standard deviation of each plotted as error bars, and a weighted line of best fit.

The line of best fit has a gradient of 3.58 and a y-intercept of 1.15m.

Total Runtime – 500.26s

Conclusion:

The final y displacement is directly proportional to the average pebble radius as predicted and shown in Figure 7. However, the y-intercept should be zero as when the average pebble radius is 0, this is the point when the final curl should also be zero.

Parameter Study – Ice Temperature:

Aim:

To investigate the effects of changing the ice temperature on the final y displacement of the curling stone.

Prediction:

The ice temperature should not affect the final curl as the thermal effects are negligible compared to the forces produced by scratches on the pebbles.

Setup:

Parameter	Ice Temperature (°C)
Minimum	-7
Maximum	-1.5
Interval	0.5
Number Of Values	12
Runs Per Value	50
Total Number Of Runs	600
Other Parameters	Kept Constant

Results:

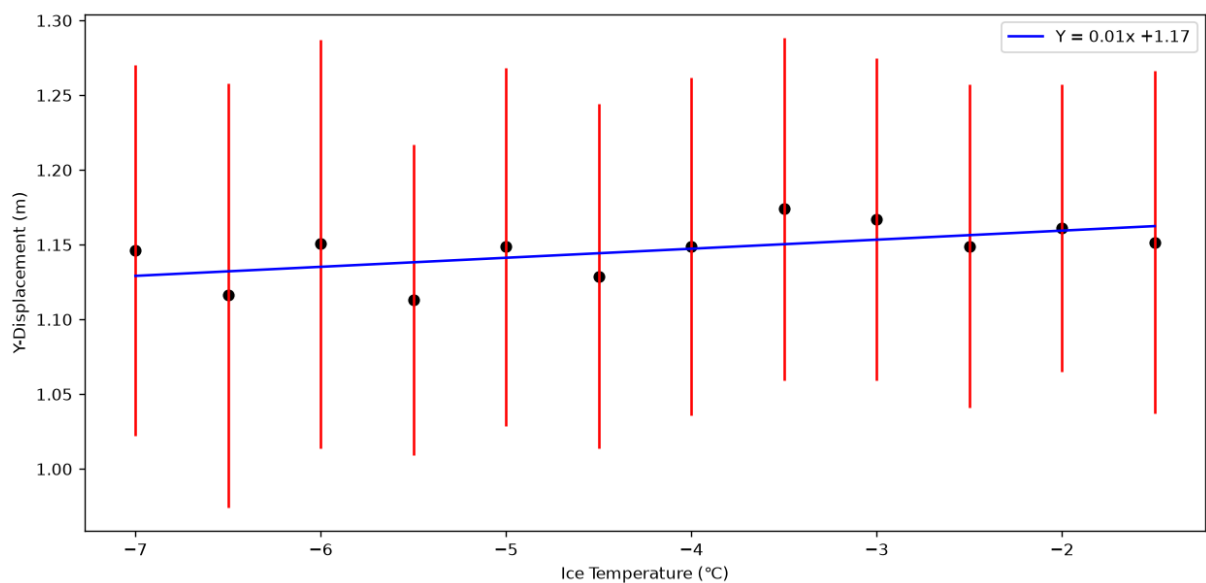


Figure 8: A graph of the final y displacement of the curling stone against the temperature of the ice, each point is the average of 50 results with the standard deviation plotted as the error bars and a weighted line of best fit.

The gradient of the line of best fit is 0.01 and the y-intercept is 1.17.

Total Runtime – 532s

Conclusion:

The final y displacement is unaffected by the temperature of the ice, shown in figure 8 and as predicted. This is shown by the gradient of the line being so close to 0.

Parameter Study – Initial Angular Velocity:

Aim:

Investigate the effects of changing the initial angular velocity on the final y displacement of the curling stone.

Prediction:

When the angular velocity is zero, there should be no y displacement. Otherwise, the final y displacement should be constant as the constant A is not currently related to the angular velocity.

Setup:

Parameter	Initial Angular Velocity (rad/s)
Minimum	0
Maximum	7
Interval	0.5
Number Of Values	15
Runs Per Value	50
Total Number Of Runs	750
Other Parameters	Kept Constant

Results:

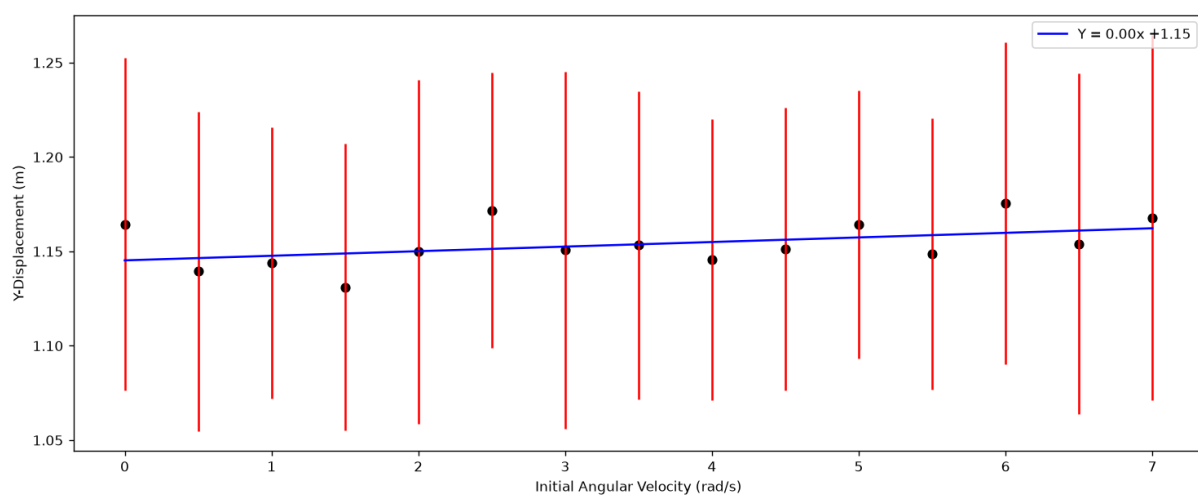


Figure 9: A graph of final y displacement of the curling stone against initial angular velocity. Each point is the average of 50 results with the standard deviation plotted as error bars, and a weighted line of best fit.

The line of best fit has a gradient of 0 and a y intercept of 1.15m.

Total Runtime – 1164s

Conclusion:

The final y displacement of the curling stone is unaffected by the initial angular velocity of the curling stone. This is as predicted, however, the y displacement should be zero when the initial angular velocity is zero, zero angular velocity means no scratches so no curl. Therefore, the model needs fixing to represent this correctly.